

PYTHON FUNCTIONS AND EXPRESSIONS

COMPUTER SCIENCE MENTORS 61A

September 2 – September 5, 2025

1 Expressions

1. What Would Python Display?

```
>>> 3
```

```
>>> "cs61a"
```

```
>>> x = 3
>>> x
```

```
>>> x = print("cs61a")
cs61a
>>> x
```

```
>>> print(print(print("cs61a")))
```

```
>>> def f1(x):
...     return x + 1
>>> f1(3)
```

```
>>> f1(2) + f1(2 + 3)
```

```
>>> def f2(y):
...     return y / 0
>>> f2(4)
```

```

>>> def f3(x, y):
...     if x > y:
...         return x
...     elif x == y:
...         return x + y
...     else:
...         return y
>>> f3(1, 2)

>>> f3(5, 5)

>>> 1 or 2 or 3

>>> 1 or 0 or 3

>>> 4 and (2 or 1/0)

>>> 0 or (not 1 and 3)

>>> (2 or 1/0) and (False or (True and (0 or 1)))

```

2. What would Python display for each expression?

- (a) `3 + 4`
- (b) `'3' + 4`
- (c) `'3 + 4'`
- (d) `'3' + '4'`

2 Functions

1. For the following expressions, simplify the operands in the order of evaluation of the entire expression

Example: `add(3, mul(4, 5))`

Order of Evaluation: `add(3, mul(4, 5))` $\rightarrow$ `add(3, 20)` $\rightarrow$ 23

- (a) `add(1, mul(2, 3))`

(b) `add(mul(2, 3), add(1, 4))`

(c) `max(mul(1, 2), add(5, 6), 3, mul(mul(3, 4), 1), 7)`

2. Write a function `fahrenheitToCelsius`, which takes in an integer `f` and returns the temperature in Celsius. Hint: the formula for Fahrenheit to Celsius is to subtract 32 and then multiply by 5/9.

```
def fahrenheitToCelsius(f):  
    """  
    >>> fahrenheitToCelsius(32)  
    0.0  
    >>> fahrenheitToCelsius(104)  
    40.0  
    """
```

3. Sally really wants to know what she needs to score to pass her class! Fill out the function `score_needed` that takes in *current_grade*, her current grade, *num_exams*, the number of exams taken so far (not including her final exam), and *pass_grade*, the grade needed to pass. Assume that every exam, including the final exam, has the same weight.

```
def score_needed(current_grade, num_exams, pass_grade):  
    """  
    >>> score_needed(89, 4, 90)  
    94  
    >>> score_needed(65, 2, 70)  
    80  
    >>> score_needed(77, 10, 78)  
    88  
    """
```

4. What would Python display?

```
>>> from operator import sub, mul
>>> def print_sub(x, y):
...     print('sub')
...     return sub(x, y)
>>> def print_mul(x, y):
...     print('mul')
...     return mul(x, y)
>>> print_sub(print_mul(505, 4), 3)
```